

Computer Science & IT Engineering

DPP 4; Top 250 Questions of DBMS Part-4



DBMS : DPP 4

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[MCQ]



Consider the following two sets of functional dependencies

$X = \{P \rightarrow Q, Q \rightarrow R, R \rightarrow P, P \rightarrow R, R \rightarrow Q, Q \rightarrow P\}$

$Y = \{P \rightarrow Q, Q \rightarrow R, R \rightarrow P\}$

Which of the following is true?

- A. $X \subset Y$
- B. $Y \subset X$
- C. $X \equiv Y$
- D. $X \neq Y$

$$X = \{P \rightarrow Q, Q \rightarrow R, R \rightarrow P, P \rightarrow R, R \rightarrow Q, Q \rightarrow P\}$$

$$Y = \{P \rightarrow Q, Q \rightarrow R, R \rightarrow P\}$$



[MSQ]



Consider the following FD sets:

$$S_1 = \{P \rightarrow R, PR \rightarrow S, T \rightarrow PS, T \rightarrow U\}$$

$$S_2 = \{P \rightarrow S, QR \rightarrow PS, R \rightarrow Q, T \rightarrow P, T \rightarrow S, T \rightarrow U\}$$

$$S_3 = \{P \rightarrow S, R \rightarrow P, R \rightarrow Q, T \rightarrow PU\}$$

Which of the following sets is equivalent?

A.

$$S_1 \equiv S_2$$

B.

$$S_2 \equiv S_3$$

C.

$$S_1 \equiv S_3$$

D.

$$S_1 \equiv S_2 \equiv S_3$$

$$S_1 = \{P \rightarrow R, PR \rightarrow S, T \rightarrow PS, T \rightarrow U\}$$

$$S_2 = \{P \rightarrow S, QR \rightarrow PS, R \rightarrow Q, T \rightarrow P, T \rightarrow S, T \rightarrow U\}$$

$$S_3 = \{P \rightarrow S, R \rightarrow P, R \rightarrow Q, T \rightarrow PU\}$$



$$S_1 = \{P \rightarrow R, PR \rightarrow S, T \rightarrow PS, T \rightarrow U\}$$

$$S_2 = \{P \rightarrow S, QR \rightarrow PS, R \rightarrow Q, T \rightarrow P, T \rightarrow S, T \rightarrow U\}$$

$$S_3 = \{P \rightarrow S, R \rightarrow P, R \rightarrow Q, T \rightarrow PU\}$$



[NAT]



Consider a relation $R = \{P, Q, R, S, T, U, V, W\}$ with the functional dependency sets $S = \{PR \rightarrow V, S \rightarrow TV, QR \rightarrow S, RV \rightarrow QS, PRS \rightarrow Q, RT \rightarrow PV\}$

The minimum numbers of single(simplified) functional dependency in the minimal cover of F is _____?

$S = \{PR \rightarrow V, S \rightarrow TV, QR \rightarrow S, RV \rightarrow QS, PRS \rightarrow Q, RT \rightarrow PV\}$



$S = \{PR \rightarrow V, S \rightarrow TV, QR \rightarrow S, RV \rightarrow QS, PRS \rightarrow Q, RT \rightarrow PV\}$



[MCQ]



Consider a schema R (ABCDEFGHIJK) with the following functional dependencies:

$F: \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJK\}$

The decomposition of R into $R_1(ABCD)$ $R_2(DE)$ $R_3(BFGH)$ $R_4(DIJK)$ is.

- A** Lossless join & dependency preserving.
- B** Lossless join & dependency not preserving.
- C** Lossy join and dependency not preserving.
- D** Lossy join and dependency preserving.

[NAT]



Consider a relation schema R with A,B,C,D,E,F with the functional dependencies:

$A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$, $E \rightarrow F$

The number of following decomposition does not gives lossless join is_____.

D₁: R₁(AB) R₂(BC) R₃(CDE) R₄(EF)

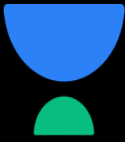
D₂: R₁(AE) R₂(BE) R₃(CDF)

D₃: R₁(ABC) R₂(CD) R₃(DEF)

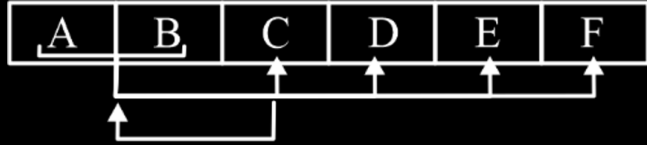
D₄: R₁(AB) R₂(BCD) R₃(EF)

D₅: R₁(ABD) R₂(BEF) R₃(EC)

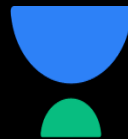
[MCQ]



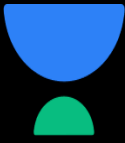
Consider the following table. The table R(ABCDEF) is in which highest normal Form?



- A** First Normal form but Not second Normal Form.
- B** Second Normal Form but Not Third Normal Form.
- C** Third Normal Form but Not BCNF
- D** Third Normal Form and BCNF



Video Solution of DPP 4



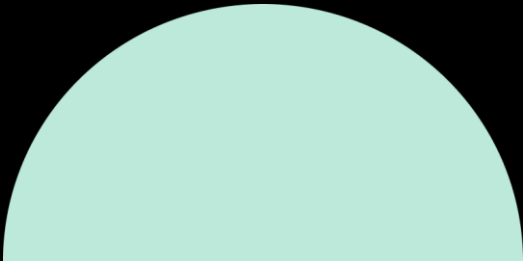
Link: <https://unacademy.com/a/Top-250-Questions-on-DBMS.html%27>

Video Solution of DPP 4 LINK



THANK YOU!

Here's to a cracking journey ahead!

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